

The Cone Theorem in the Kähler MMP

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Big picture

The cone theorem locates the closed cone of curves' negative directions and shows they are cut out by rational curves of bounded degree — it is the entry point to the minimal model program, since it names exactly the extremal rays one is allowed to contract. In the Kähler category honest curves can be scarce, so the classical Mori cone $\overline{NE}(X)$ is replaced by the **generalized Mori cone** $\overline{NA}(X)$ of positive currents (??). ?? reviews the projective cone theorem as a baseline; ??? prove it for Kähler threefolds, first for terminal X (Höring–Peternell) and then for dlt pairs (Das–Hacon); ?? upgrades this to families of threefolds, the input needed for the 4-dimensional Kähler MMP with scaling; and ?? removes the dimension restriction entirely, at the cost of assuming BDPP — now a theorem of Ou. ?? collects the applications: the contraction theorem, one step of the Kähler MMP, and what each ingredient buys in families and in arbitrary dimension.

The major references are [?], [?], [?] and [?].

The first main result is the cone theorem for Kähler threefolds.

Theorem 0.1 ([?, Theorem 6.3], [?, Theorem 1.3]). *Let (X, B) be a compact Kähler threefold dlt pair. Assume further that $K_X + B$ is pseudo-effective. Then there exists a countable collection of rational curves $\{C_i\}_{i \in I}$ such that $0 < -(K_X + B) \cdot C_i \leq 6$ for all $i \in I$ and*

$$\overline{NA}(X) = \overline{NA}(X)_{(K_X+B) \geq 0} + \sum_{i \in I} \mathbb{R}_{\geq 0} \cdot [C_i]$$

Moreover, the following holds:

- (1) *For any Kähler class ω , there are only finitely many extremal rays $R_i := \mathbb{R}_{\geq 0} \cdot [C_i]$ such that $(K_X + B + \omega) \cdot R_i < 0$.*
- (2) *For any $(K_X + B)$ -negative extremal ray $R = \mathbb{R}_{\geq 0} \cdot [C_i]$, there is a nef class $\alpha \in H_{BC}^{1,1}(X)$ such that $\alpha^\perp \cap \overline{NA}(X) = R$ and $\alpha = K_X + B + \eta$ for some Kähler class η .*

The second main result is Hacon–Păun's proof of the cone theorem in arbitrary dimension.

Theorem 0.2 ([?, Theorem 0.5]). Assume that the BDPP conjecture holds in dimension n (resp. in dimension $n - 1$) — which is true by [?, Theorem 1]. Let X be a compact $\mathbb{Q}$ -factorial Kähler variety of dimension n such that $(X, B + \beta)$ is generalized klt.

Then there are at most countably many rational curves $\{\Gamma_i\}_{i \in I}$ such that

$$\overline{\text{NA}}(X) = \overline{\text{NA}}(X)_{K_X + B + \beta_X \geq 0} + \sum_{i \in I} \mathbb{R}^+ [\Gamma_i]$$

where $0 < -(K_X + B + \beta_X) \cdot \Gamma_i \leq 2n$. Moreover, if $B + \beta_X$ (or $K_X + B + \beta_X$) is big, then I is finite.

Remark 0.3 (Kähler versus projective). Two features distinguish these statements from the projective cone theorem (??). First, the decomposition is of the *generalized* Mori cone $\overline{\text{NA}}(X)$, not the Mori cone $\overline{\text{NE}}(X)$: in the Kähler setting $\overline{\text{NE}}(X)$ can be strictly smaller than $\overline{\text{NA}}(X)$, and may even be trivial while X carries no curves at all (??), so it does not see enough of the geometry to admit a contraction theory. Second, the pseudo-effectivity of $K_X + B$ — absent from the projective statement — is essential here: it is what provides a Zariski decomposition (in ??) or a well-behaved null locus (in ??) to run bend-and-break. The length bound itself is unchanged: $0 < -(K_X + B) \cdot \Gamma_i \leq 2n$ is the one-sided form of Kawamata's $-2n \leq (K_X + B) \cdot C < 0$.

⚠ Warning. The pseudo-effectivity of $K_X + B$ in ?? and ?? is **not a technical convenience** — it **cannot be dropped**: it is exactly what supplies the Zariski decomposition used to run bend-and-break in dimension three (??), and the well-behaved null locus used in arbitrary dimension (??). Without it there is no reason for the cone to be generated by countably many bounded-length rays at all.

1 A brief review of the cone theorem for projective varieties

Let us first give a brief review of the standard cone theorem that we are familiar with.

Theorem 1.1 ([?, Theorem 6.4]; cf. [?, Theorem 7.1]). Let $(X/Z, B)$ be a projective klt pair of dimension d .

(1) There are countably many rational curves $C_i \subset X$ such that

$$\overline{\text{NE}}(X/Z) = \overline{\text{NE}}(X/Z)_{K_X + B \geq 0} + \sum_i \mathbb{R}^+ [C_i],$$

where the negative extremal rays R_i are all spanned by the curve class $[C_i]$.

(2) Each R_i is generated by the class of some rational curve C_i satisfying

$$-2d \leq (K_X + B) \cdot C_i < 0.$$

(3) If $F \subset \overline{NE}(X)$ is a $(K_X + B)$ -negative extremal face, then there exists a unique contraction morphism

$$\text{cont}_F: X \rightarrow Z$$

such that a curve $C \subset X$ is contracted to a point if and only if $[C] \in F$.

(4) Let L be a Cartier divisor on X such that $L \cdot R = 0$ for some $(K_X + B)$ -negative extremal ray R . Then there is a Cartier divisor L_Y on Y such that

$$L \sim (\text{cont}_R)^* L_Y.$$

We refer to (2) as **Kawamata's length of extremal rays**. Note that here B is allowed to be an $\mathbb{R}$ -divisor.

💡 Idea of proof. Five steps: extract the length of extremal rays from Mori's bend-and-break; perturb B to a $\mathbb{Q}$ -divisor so the rationality theorem applies; attach a contraction to the rational supporting hyperplane of each candidate face; show a Hahn–Banach separation argument forces these faces to generate the whole cone; and finally use the rationality theorem's denominator bound to show only finitely many rays are negative against any fixed ample perturbation.

Proof. We argue by induction on $\rho(X/S) = \dim N_1(X/S)$; the relative setting is essential throughout.

Step 1 (Length of extremal rays). For every $(K_X + B)$ -negative extremal ray R there is a rational curve C with $[C] \in R$ and $-2d \leq (K_X + B) \cdot C < 0$. This is Kawamata's bound, obtained from Mori's bend-and-break [?, Cor. 2.8.4, Thm. 7.1]; it proves (2) and provides the boundedness used in Step 5.

Step 2 (Reduction to a $\mathbb{Q}$ -divisor). A priori $K_X + B$ is only an $\mathbb{R}$ -divisor. Fix a relatively ample divisor A and $\epsilon \in \mathbb{Q}_{>0}$. Write $K_X + B = \sum_{i=1}^t r_i D_i$ with D_i Cartier, and approximate the r_i by rationals r'_i so that $\sum_i (r_i - r'_i) D_i + \frac{\epsilon}{3} A$ is relatively ample. Set $B' = \sum_i r'_i D_i - K_X$ and write $B' = (B')^+ - (B')^-$ with $(B')^\pm$ effective $\mathbb{Q}$ -divisors without common component. For $r'_i - r_i$ small, the coefficients of $(B')^-$ are small, so there is an effective $\mathbb{Q}$ -divisor B'' with small coefficients and $\frac{\epsilon}{3} A - (B')^- \sim_{\mathbb{Q}} B''$; shrinking further, $(X, (B')^+ + B'')$ is again klt. For every curve C ,

$$C \cdot (K_X + (B')^+ + B'' + \frac{\epsilon}{3} A) = C \cdot (K_X + B + \epsilon A) - C \cdot (B - B' + \frac{1}{3} \epsilon A) \leq C \cdot (K_X + B + \epsilon A),$$

so $\overline{NE}(X/S)_{K_X + (B')^+ + B'' + \frac{\epsilon}{3} A \geq 0} \subset \overline{NE}(X/S)_{K_X + B + \epsilon A \geq 0}$, and the assertion for (X, B) follows once it is known for the klt $\mathbb{Q}$ -pair $(X, (B')^+ + B'')$. Hence we may assume B is a $\mathbb{Q}$ -divisor, as required to apply the rationality theorem.

Step 3 (Contraction attached to a supporting plane). If $\rho(X/S) = 1$ there is nothing to prove, so assume $\rho(X/S) > 1$ and that $K_X + B$ is not relatively nef. For any relatively ample $\mathbb{Q}$ -divisor H , the rationality theorem yields a rational nef threshold

$$r_H = \max\{t \in \mathbb{R} \mid H + t(K_X + B) \text{ is relatively nef}\} \in \mathbb{Q}, \quad L_H := H + r_H(K_X + B).$$

Then L_H is relatively nef but not relatively ample, so $F_H := \overline{\text{NE}}(X/S)_{L_H=0}$ is a face: if $\alpha_1 + \alpha_2 \in F_H$ then, by nefness and the duality $\text{Nef} = \overline{\text{NE}}^*$ (??), $\alpha_i \cdot L_H \geq 0$ forces $\alpha_1, \alpha_2 \in F_H$. Being a $(K_X + B)$ -negative extremal face with a rational supporting function, F_H admits a contraction $h_H: X \rightarrow Y_H$.

Step 4 (The cone is generated by these faces). Let $\mathcal{C}$ be the smallest closed convex cone containing $\overline{\text{NE}}(X/S)_{K_X+B \geq 0}$ and all F_H with $L_H \not\equiv_S 0$, i.e. $\mathcal{C} = \overline{\text{NE}}(X/S)_{K_X+B \geq 0} + \sum_{L_H \not\equiv_S 0} F_H$. We claim $\mathcal{C} = \overline{\text{NE}}(X/S)$. If not, by Hahn–Banach there is a $\mathbb{Q}$ -divisor M with $M \cdot v > 0$ on $\mathcal{C} \setminus \{0\}$ and $M \cdot v_0 < 0$ for some $v_0 \in \overline{\text{NE}}(X/S)$. Then $M \not\equiv_S t(K_X + B)$: for if $M \equiv_S t(K_X + B)$ then $t < 0$ (both M and $-(K_X + B)$ are positive on some $F_H \setminus \{0\}$), while $M \cdot v_0 < 0$ forces $v_0 \in \overline{\text{NE}}(X/S)_{K_X+B \geq 0} \setminus \{0\} \subset \mathcal{C} \setminus \{0\}$, contradicting $M > 0$ there. Now the dual of $\overline{\text{NE}}(X/S)_{K_X+B \geq 0}$ is $\text{Amp}(X/S) + \mathbb{R}_{\geq 0}(K_X + B)$ (dual of a half-space is a ray; dual of the Mori cone is the nef cone, ??), and M , being positive on $\overline{\text{NE}}(X/S)_{K_X+B \geq 0} \setminus \{0\}$, is interior in it; hence $M = H + t(K_X + B)$ with H relatively ample and $t \in \mathbb{Q}_{>0}$. As M is not relatively nef, $t > r_H$. But $L_H \not\equiv_S 0$, so $F_H \subset \mathcal{C}$ and $M > 0$ on $F_H \setminus \{0\}$; taking C with $[C] \in F_H$,

$$C \cdot (H + t(K_X + B)) = \underbrace{C \cdot (H + r_H(K_X + B))}_{=0} + (t - r_H) C \cdot (K_X + B) > 0$$

forces $C \cdot (K_X + B) > 0$, impossible on a negative face. This proves $\mathcal{C} = \overline{\text{NE}}(X/S)$, which is (1). Part (3) is the contraction of Step 3, and part (4) is the induced descent of L via the base-point-free theorem [?, Thm. 7.1].

Step 5 (Discreteness). Only finitely many R_i are $(K_X + B + \epsilon A)$ -negative. For each such ray take the contraction $h_i: X \rightarrow Y_i$ and the generator $v_i \in R_i$ with $a(K_X + B) \cdot v_i = -1$. Complete $a(K_X + B)$ to a basis by relatively ample Cartier $H_1, \dots, H_{\rho-1}$; since $\dim N_1(X/Y_i) = 1$, the rationality theorem bounds the denominators $q_{ij} \leq a(b+1)$ in $r_{ij}/a = p_{ij}/q_{ij}$, so $(a(b+1))!(H_j \cdot v_i) \in \mathbb{Z}$. For $N \gg 0$ with $NA - H_j$ relatively ample and $(K_X + B + \epsilon A) \cdot v_i < 0$, one has $H_j \cdot v_i < NA \cdot v_i < N/(a\epsilon)$, so $H_j \cdot v_i$ takes finitely many values, whence finitely many rays. (Equivalently, by Step 1 the curves C_i with $(K_X + B + \epsilon A) \cdot C_i < 0$ satisfy $A \cdot C_i < 2b/\epsilon$, forming a bounded family with finitely many classes.) Letting $\epsilon \rightarrow 0$ gives countability of all the R_i . ■

2 The generalized Mori cone and the Mori cone

Before entering the Kähler setting we fix the cones that replace the classical Mori cone. Throughout, X is a normal compact Kähler variety and classes live in $N_1(X) \subset H_{\text{BC}}^{1,1}(X)^\vee$.

Definition 2.1 ([?, p. 226], [?, Def. 2.2]). We define the generalized Mori cone $\overline{NA}(X) \subset N_1(X)$ to be the closed cone generated by the classes of positive closed $(1, 1)$ -currents (via the duality pairing between $H_{BC}^{1,1}(X)$ and $H_{1,1}(X, \mathbb{R})$).

We define the **Mori cone**

$$\overline{NE}(X) \subset \overline{NA}(X)$$

as the closure of the cone generated by the currents of integration T_C , where $C \subset X$ is an irreducible curve.

 **Intuition.** Think of $\overline{NA}(X)$ as the closed convex hull of every way a positive current can be swept out on X — the right receptacle for "effective classes" once honest curves are scarce, as happens in the Kähler category.

The cone $\overline{NA}(X)$ serves as a substitute for the classical Mori cone in the Kähler setting. For an $\mathbb{R}$ -Cartier class D we write, as usual, $\overline{NA}(X)_{D \geq 0} = \overline{NA}(X) \cap \{\gamma : D \cdot \gamma \geq 0\}$, and similarly $\overline{NE}(X)_{D \geq 0}$.

Remark 2.2. In the complex-analytic setting it may happen that

$$\overline{NE}(X) \subsetneq \overline{NA}(X).$$

For instance, there exist K3 surfaces with Picard number 0, carrying no curve C at all, yet many positive closed currents.

The inclusion can be strict even for projective manifolds. For projective K3 surfaces, $\rho(X) \leq 20 = h^{1,1}(X)$ by the Torelli theorem, so $\overline{NE}(X)$ lies in the $\rho(X)$ -dimensional subspace $N_1(X) \subsetneq H_{1,1}(X, \mathbb{R})$, while $\overline{NA}(X)$ can contain classes outside $N_1(X)$, witnessing the strict inclusion.

The two facts about these cones that drive the Kähler cone theorem are a duality statement and a Kählerness (Kleiman-type) criterion.

Proposition 2.3 (Duality; [?, Prop. 3.9], [?, Cor. 0.4], [?, Lem. 2.3]). *Let X be a normal compact Kähler variety with rational singularities. Then $\text{Nef}(X)$ and $\overline{NA}(X)$ are dual to each other via the canonical isomorphism $N^1(X) \rightarrow N_1(X)^*$. In particular, a class $\omega \in H_{BC}^{1,1}(X)$ is Kähler if and only if $\omega \cdot \gamma > 0$ for all $\gamma \in \overline{NA}(X) \setminus \{0\}$.*

Proposition 2.4 (Kählerness criterion for threefolds; [?, Cor. 3.16, Thm. 3.17–3.18], [?, Thm. 3.7]). *Let X be a normal compact Kähler 3-fold with rational singularities such that every irreducible curve $C \subset X$ has $[C] \neq 0$ in $N_1(X)$, and let $[\eta] \in N^1(X)$. Then $[\eta]$ is a Kähler class if and only if $[\eta]$ is strictly positive on $\overline{NA}(X) \setminus \{0\}$.*

We shall use $\overline{\text{NA}}$ freely. We also record the Demailly–Păun description of the generators of $\overline{\text{NA}}$ on a smooth model, which is the engine that upgrades the Mori cone statement to the generalized one.

Fact 2.5 (Demailly–Păun decomposition; [?, Cor. 0.4]). Let $\hat{X}$ be a compact Kähler manifold. Then $\overline{\text{NA}}(\hat{X})$ is the closure of the convex cone generated by the classes $[\hat{\omega}]^2$, $[\hat{\omega}] \cdot [\hat{S}]$ and $[\hat{C}]$, where $\hat{\omega}$ is a Kähler form, $\hat{S}$ an irreducible surface and $\hat{C}$ an irreducible curve on $\hat{X}$.

3 Höring–Peternell's approach for Kähler 3-folds

Throughout this section X is a normal $\mathbb{Q}$ -factorial compact Kähler threefold with at most terminal singularities and K_X pseudo-effective. Let $K_X = P + N$ be the Zariski decomposition with negative part $N = \sum a_j S_j$, and set

$$b := \max\{1, -K_X \cdot Z \mid Z \text{ a curve with } Z \subset S_{j,\text{sing}} \text{ or } Z \subset S_j \cap S_{j'}\}, \quad d := \max\{3, b\}.$$

3.1 Mori bend and break for Kähler threefolds

The heart of the argument is that curves of large anti-canonical degree break.

Theorem 3.1 (Mori bend and break; [?, Cor. 5.7, Lem. 5.8]). *With X and d as above:*

- (a) (Breaking.) *If $C \subset X$ is a curve with $-K_X \cdot C > d$, then $[C] = [C_1] + [C_2]$ for non-zero effective (integral) 1-cycles C_1, C_2 on X .*
- (b) (Rational curves.) *Let $\mathbb{R}^+[\Gamma_i]$ be a K_X -negative extremal ray of $\overline{\text{NE}}(X)$ with Γ_i not very rigid. Then there is a curve C with $[C] \in \mathbb{R}^+[\Gamma_i]$ and $\dim_C \text{Chow}(X) > 0$, and there is a rational curve C with $[C] \in \mathbb{R}^+[\Gamma_i]$.*

Idea of proof. Large anti-canonical degree forces the curve to deform; its deformations sweep out a uniruled surface from the Zariski decomposition of K_X , and ordinary surface bend-and-break on that uniruled surface produces the splitting, or a rational curve, after pushing forward.

Proof. (a) *Breaking.* Since $-K_X \cdot C > d \geq b$, the curve C is deformable: $\dim_C \text{Chow}(X) > 0$ (a curve of sufficiently large anti-canonical degree on a Kähler 3-fold moves, [?, §5]). Let $(C_t)_{t \in T}$ be a family of deformations. By [?, Lem. 5.4] there is a unique Zariski surface S_j containing all C_t , with

$$K_{S_j} \cdot C < K_X \cdot C < -d.$$

By definition of b we have $C \not\subset S_{j,\text{sing}}$ (otherwise $-K_X \cdot C \leq b \leq d$, contradicting $-K_X \cdot C > d$). Let $\pi_j: \hat{S}_j \rightarrow S_j$ be the composition of normalization and minimal

resolution; the strict transform $\hat{C}$ of C is well defined and

$$K_{\hat{S}_j} \cdot \hat{C} \leq K_{S_j} \cdot C < -3.$$

Since K_{S_j} is not pseudo-effective, $\hat{S}_j$ is a smooth uniruled surface, so Mori bend-and-break on surfaces gives an effective 1-cycle $\sum_{k=1}^m C_k$ with $m \geq 2$, $[\sum_k C_k] = [\hat{C}]$, and $K_{\hat{S}_j} \cdot C_1 < 0$, $K_{\hat{S}_j} \cdot C_2 < 0$. As $K_{\hat{S}_j}$ is π_j -nef (minimal resolution), any C_k with $(\pi_j)_* C_k = 0$ would be π_j -vertical, hence $K_{\hat{S}_j} \cdot C_k \geq 0$; thus $(\pi_j)_* C_1 \neq 0$ and $(\pi_j)_* C_2 \neq 0$. Therefore

$$[C] = (\pi_j)_* [\hat{C}] = \sum_{k=1}^m (\pi_j)_* [C_k],$$

with the first two summands non-zero, so $[C]$ splits as $[C'_1] + [C'_2]$.

(b) *Rational curves in an extremal ray.* Let $R = \mathbb{R}^+[\Gamma_i]$ be extremal with Γ_i not very rigid, and set

$$m := \min\{k \in \mathbb{N} \mid \exists C \subset X, [C] \in R, \dim_{kC} \text{Chow}(X) > 0\}.$$

We show $m = 1$. If $m > 1$, choose C realizing m with deformation family $(C_t)_{t \in T}$; by minimality of m the family has no fixed component, so a general C_t has every irreducible component C'_t with $\dim_{C'_t} \text{Chow}(X) > 0$, and extremality gives $[C'_t] \in R$ — contradicting minimality. Hence there is C with $[C] \in R$ and $\dim_C \text{Chow}(X) > 0$, which is the first assertion of (b). For the rational curve, by [?, Lem. 5.4] the deformations of C lie in some Zariski surface S_j with $K_{S_j} \cdot C < K_X \cdot C < 0$; after a general deformation $C \not\subset S_{j,\text{sing}}$, and on $\hat{S}_j$ the strict transform satisfies $K_{\hat{S}_j} \cdot \hat{C} \leq K_{S_j} \cdot C < 0$. Surface bend-and-break yields an effective $\mathbb{Q}^+$ -cycle $\sum_k \alpha_k C_k$ with $[\sum_k \alpha_k C_k] = [\hat{C}]$ and C_1 a rational curve with $K_{\hat{S}_j} \cdot C_1 < 0$; as $K_{\hat{S}_j}$ is π_j -nef, $(\pi_j)_* C_1 \neq 0$, so

$$[C] = \sum_k \alpha_k (\pi_j)_* [C_k], \quad \alpha_1 (\pi_j)_* [C_1] \neq 0.$$

By extremality, $[C_1]$ — the class of a rational curve — is a positive multiple of $[\Gamma_i]$, so R contains the class of a rational curve. ■

Remark 3.2. In the language of [?, Def. 4.5, Prop. 4.6], the theorem says that $K_X + \Delta$ has the **bend-and-break property**: there is $d \in \mathbb{Q}^+$ such that every curve C with $-(K_X + \Delta) \cdot C > d$ splits as $[C] = [C_1] + [C_2]$. This is what makes Kawamata's length effective.

3.2 The cone theorem for Kähler 3-folds

Theorem 3.3 ([?, Theorem 6.3]). *Let X be a normal $\mathbb{Q}$ -factorial compact Kähler threefold with at most terminal singularities such that K_X is pseudo-effective. Then there exist $d \in \mathbb{N}$ and a countable family $(\Gamma_i)_{i \in I}$ of curves on X such that*

$$0 < -K_X \cdot \Gamma_i \leq d \quad \text{and} \quad \overline{\text{NA}}(X) = \overline{\text{NA}}(X)_{K_X \geq 0} + \sum_{i \in I} \mathbb{R}^+ [\Gamma_i].$$

Moreover, if the ray $\mathbb{R}^+ [\Gamma_i]$ is extremal in $\overline{\text{NA}}(X)$, there exists a rational curve C_i on X with $[C_i] \in \mathbb{R}^+ [\Gamma_i]$.

The proof splits into a statement for the Mori cone and an upgrade to the generalized Mori cone. We first record the tool that gives closedness.

Proposition 3.4 (Closedness under bounded length; [?, Lem. 6.1]). *Let $N \subset \overline{\text{NA}}(X) \subset N_1(X)$ be a closed convex cone, and suppose there are $d \in \mathbb{N}$ and a countable family $(\Gamma_i)_{i \in I}$ with $0 < -K_X \cdot \Gamma_i \leq d$ and*

$$N = \overline{N_{K_X \geq 0} + \sum_{i \in I} \mathbb{R}^+ [\Gamma_i]}.$$

Then $N = N_{K_X \geq 0} + \sum_{i \in I} \mathbb{R}^+ [\Gamma_i]$. The same holds with a dlt or generalized log canonical class in place of K_X .

Proof. Set $V := N_{K_X \geq 0} + \sum_i \mathbb{R}^+ [\Gamma_i]$; we must show $\overline{V} = V$. Since $V \subset \overline{\text{NA}}(X)$, V contains no line and is the convex hull of its extremal rays, so it suffices to show that every extremal ray $\mathbb{R}^+ [r]$ of $\overline{V}$ with $K_X \cdot r < 0$ lies in V . Fix a Kähler form ω and $\varepsilon > 0$ with $(K_X + \varepsilon\omega) \cdot r < 0$. Then

$$N = \overline{N_{K_X + \varepsilon\omega \geq 0} + \sum_{j \in J} \mathbb{R}^+ [\Gamma_j]},$$

the sum over the $j \in I$ with $(K_X + \varepsilon\omega) \cdot \Gamma_j < 0$; for these, $-K_X \cdot \Gamma_j \leq d$ gives $\omega \cdot \Gamma_j \leq d/\varepsilon$, so by boundedness of the ω -degree there are only finitely many classes, say $J = \{1, \dots, q\}$, and $N_{K_X + \varepsilon\omega \geq 0} + \sum_{j=1}^q \mathbb{R}^+ [\Gamma_j]$ is already closed.

Write $r = \lim_m (r_m + s_m)$ with $(K_X + \varepsilon\omega) \cdot r_m \geq 0$ and $s_m = \sum_{j=1}^q \lambda_{j,m} \Gamma_j$. As $\omega \cdot (r_m + s_m) \rightarrow \omega \cdot r$ and $\omega \geq 0$ on N , the sequences $\omega \cdot r_m$ and $\omega \cdot \lambda_{j,m} \Gamma_j$ are bounded, so after passing to a subsequence $r_m \rightarrow r_\infty$ and $\lambda_{j,m} \Gamma_j \rightarrow \mu_j \Gamma_j$. Then $r = r_\infty + \sum_j \mu_j \Gamma_j$ with all summands in $\overline{V}$; since r is extremal in $\overline{V}$, each is a non-negative multiple of r . But $(K_X + \varepsilon\omega) \cdot r_\infty \geq 0$ while $(K_X + \varepsilon\omega) \cdot r < 0$ forces $r_\infty = 0$, so some $\mu_{j_0} \Gamma_{j_0}$ is a positive multiple of r ; hence $\mathbb{R}^+ [r] = \mathbb{R}^+ [\Gamma_{j_0}] \subset V$. ■

Proposition 3.5 (Weak cone theorem, Mori cone version; [?, Theorem 6.2]). *With X as in*

??, there are $d \in \mathbb{N}$ and a countable family $(\Gamma_i)_{i \in I}$ with $0 < -K_X \cdot \Gamma_i \leq d$ and

$$\overline{\text{NE}}(X) = \overline{\text{NE}}(X)_{K_X \geq 0} + \sum_{i \in I} \mathbb{R}^+[\Gamma_i],$$

and every extremal ray of $\overline{\text{NE}}(X)$ contains the class of a rational curve.

Proof. Take $d = \max\{3, b\}$ as above. There are at most countably many curve classes in $\overline{\text{NE}}(X)$ (integral curves are lattice points), and we pick a representative Γ_i of each class with $0 < -K_X \cdot \Gamma_i \leq d$. Set

$$V := \overline{\text{NE}}(X)_{K_X \geq 0} + \sum_{0 < -K_X \cdot \Gamma_i \leq d} \mathbb{R}^+[\Gamma_i],$$

which is closed by ??. Fix a Kähler form ω with $\omega \cdot C \geq 1$ for every irreducible curve C (possible after rescaling, since only finitely many classes have ω -degree < 1); the same then holds for effective 1-cycles.

By ?? it suffices to show $\overline{\text{NE}}(X) = \overline{V}$, i.e. $[C] \in V$ for every irreducible curve C . We induct on $\ell := \omega \cdot C$. The case $\ell = 0$ is trivial. Assume the claim for all curves of degree $\leq \ell - 1$, and let $\ell - 1 < \omega \cdot C \leq \ell$. If $-K_X \cdot C \leq d$ we are done. Otherwise ??(a) gives $[C] = [C_1] + [C_2]$ with C_i effective 1-cycles; since $\omega \cdot C_i \geq 1$ we get $\omega \cdot C_i \leq \ell - 1$, so by induction $[C_1], [C_2] \in V$ and hence $[C] \in V$.

Finally, if $\mathbb{R}^+[\Gamma_i]$ is extremal, then ??(b) (together with the fact that a very rigid K_X -negative curve is already a rational curve of bounded degree) produces a rational curve in the ray. ■

Proof of ??. Set $V := \overline{\text{NA}}(X)_{K_X \geq 0} + \sum_{i \in I} \mathbb{R}^+[\Gamma_i]$ with the Γ_i from ??. By ?? it suffices to prove $\overline{\text{NA}}(X) \subset \overline{V}$.

Let $\pi: \hat{X} \rightarrow X$ be a desingularization and $\alpha \in \overline{\text{NA}}(X)$. By the Kähler–Mori push-forward, $\alpha = \pi_*(\hat{\alpha})$ for some $\hat{\alpha} \in \overline{\text{NA}}(\hat{X})$. By ??, $\overline{\text{NA}}(\hat{X})$ is generated by classes of the form $[\hat{\omega}]^2$, $[\hat{\omega}] \cdot [\hat{S}]$ and $[\hat{C}]$; we show the π -image of each type lies in $\overline{V}$.

Case $\hat{\alpha} = [\hat{\omega}]^2$. Since π^*K_X is pseudo-effective, $K_X \cdot \alpha = \pi^*K_X \cdot \hat{\alpha} \geq 0$, so $\alpha \in \overline{\text{NA}}(X)_{K_X \geq 0} \subset \overline{V}$.

Case $\hat{\alpha} = [\hat{C}]$. Put $C = \pi_*\hat{C}$, so $\alpha = [C]$. Under the inclusion $\overline{\text{NE}}(X)_{K_X \geq 0} + \sum \mathbb{R}^+[\Gamma_i] \hookrightarrow \overline{\text{NA}}(X)_{K_X \geq 0} + \sum \mathbb{R}^+[\Gamma_i]$, ?? gives $[C] \in V$.

Case $\hat{\alpha} = [\hat{\omega}] \cdot [\hat{S}]$. If $\pi^*K_X \cdot [\hat{\omega}] \cdot [\hat{S}] \geq 0$ then $\alpha \in \overline{\text{NA}}(X)_{K_X \geq 0}$. Suppose instead $\pi^*K_X \cdot [\hat{\omega}] \cdot [\hat{S}] < 0$. We claim $\hat{\alpha} \in \overline{\text{NE}}(\hat{X})$; since $\pi_*\overline{\text{NE}}(\hat{X}) = \overline{\text{NE}}(X)$, this reduces to the curve case. By the projection formula $\pi(\hat{S})$ is not a point, and as X has isolated singularities $\pi(\hat{S})$ is a surface S . The restriction $K_X|_S$ is not pseudo-effective, so S is one of the Zariski surfaces S_j ; in particular S is uniruled and so is $\hat{S}$. Let $\pi_j: \hat{S}_j \rightarrow \hat{S} \subset \hat{X}$ be the composition of normalization and minimal resolution; then $\hat{S}_j$ is uniruled and projective, whence $H^2(\hat{S}_j, \mathcal{O}) = 0$ and the Chern class map $\text{Pic } \hat{S}_j \rightarrow H^2(\hat{S}_j, \mathbb{Z})$ is surjective,

so $\text{NS}_{\mathbb{R}}(\hat{S}_j) = H^{1,1}(\hat{S}_j) \cap H^2(\hat{S}_j, \mathbb{R})$. Thus $[\pi_j^* \hat{\omega}]$ is a nef and big $\mathbb{R}$ -divisor class, a limit of ample $\mathbb{Q}$ -divisor classes H_m represented by effective $\mathbb{Q}$ -1-cycles $[C_m]$. One checks $(\pi_j)_*[C_m] \rightarrow \hat{\alpha}$: by duality it suffices that for every real closed $(1, 1)$ -form η ,

$$[\eta] \cdot (\pi_j)_*[C_m] = [\pi_j^* \eta] \cdot [C_m] \longrightarrow \int_{\hat{S}_j} \pi_j^* \eta \wedge \pi_j^* \omega = [\eta] \cdot [\hat{\omega}] \cdot [\hat{S}],$$

which follows from the construction of $[C_m]$. Hence $\hat{\alpha} \in \overline{\text{NE}}(\hat{X})$ and $\alpha \in \overline{V}$. ■

4 Das–Hacon’s approach for Kähler threefold dlt pairs

The result of the previous section holds more generally for dlt pairs; we omit the terminal case and prove the dlt version, ???. This is [?, Theorem 1.3] (see also [?, Thm. 2.22]).

💡 Idea of proof. The strategy follows ???: Boucksom’s nefness criterion [?] reconciles metric and algebraic nefness of $K_X + B$; Kähler threefold dlt pairs satisfy Mori bend-and-break (??), giving Kawamata’s length; and the closedness ?? together with the duality ?? reduces the generalized cone to the Mori cone via the Demailly–Păun generators (??). The one genuinely new feature is the exposed-point argument, which replaces the induction on ω -degree used in ??.

Claim 4.1 (Bend and break for dlt pairs; [?, Claim 4.8]). *There is $d > 0$ such that if $C \subset X$ is a curve with $-(K_X + B + \omega) \cdot C > d$, then $[C] = [C_1] + [C_2]$ with C_1, C_2 non-zero integral effective 1-cycles.*

The following lemma is pure cone geometry and encapsulates the closedness input; we omit its proof.

Lemma 4.2 (Cone geometry / closedness). *Let $V := \overline{\text{NA}}(X)_{(K_X+B) \geq 0} + \sum_{i \in I} \mathbb{R}^+[\Gamma_i]$, where the Γ_i are rational curves with $0 < -(K_X + B) \cdot \Gamma_i \leq d$. Then V is a closed convex cone. In particular $\overline{\text{NA}}(X) = V$ if and only if $\overline{\text{NA}}(X) = \overline{V}$; equivalently, it suffices to show that every generator of $\overline{\text{NA}}(X)$ lies in $\overline{V}$.*

Proof of ???. *Reduction to strongly $\mathbb{Q}$ -factorial X .* By standard arguments (cf. [?, Cor. 5.3]), passing to a strongly $\mathbb{Q}$ -factorial model (which exists by [?, Lem. 2.28]) we may assume X is strongly $\mathbb{Q}$ -factorial.

Extremal rays are spanned by short rational curves. Let K be a compact convex slice of $\overline{\text{NA}}(X)$ of Euclidean dimension $n - 1$, $n = \dim_{\mathbb{R}} H_{\text{BC}}^{1,1}(X)$ (so the cone on K is $\overline{\text{NA}}(X)$), and let R be an extremal ray corresponding to an exposed point $\{x\} \in K$. By definition there is a class $\alpha \in H_{\text{BC}}^{1,1}(X)$ with $K \subset \alpha_{\geq 0}$ and $\{x\} = K \cap \alpha^{\perp}$, so $R = \overline{\text{NA}}(X) \cap \alpha^{\perp}$. Since α is non-negative on $\overline{\text{NA}}(X)$ it is nef (??). If $(K_X + B) \cdot R < 0$, then for $0 < a \ll 1$ the class $\beta := \alpha - a(K_X + B)$

is strictly positive on $\overline{\text{NA}}(X) \setminus \{0\}$, hence Kähler; writing $\alpha = a(K_X + B + \frac{1}{a}\beta)$ and rescaling we may assume $\alpha \equiv K_X + B + \omega$ with ω Kähler. By the existence of flipping/divisorial contractions ([?, Thms. 6.2, 6.9]), $R = \mathbb{R}^+[\Gamma]$ for a curve Γ , and since $-(K_X + B)$ is relatively ample for the contraction with (X, B) dlt, Γ may be taken to be a rational curve with $0 > (K_X + B) \cdot \Gamma \geq -6$.

Countability. As X is Kähler, a Douady space argument shows there are at most countably many such extremal rays $\{\mathbb{R}^+[\Gamma_i]\}_{i \in I}$.

Reduction to $\overline{\text{NA}}(X) = \overline{V}$. Put $V := \overline{\text{NA}}(X)_{K_X+B \geq 0} + \sum_{i \in I} \mathbb{R}^+[\Gamma_i]$. Following the proof of ?? it suffices to show $\overline{\text{NA}}(X) = \overline{V}$; by ?? the point is the closedness of V . Suppose $\overline{\text{NA}}(X) \setminus \overline{V} \neq \emptyset$. Since $\overline{V}$ is convex and $\overline{\text{NA}}(X)$ is generated by K , there is an exposed point $x \in K$ not in $\overline{V}$; as V contains everything non-negative against $K_X + B$, the ray R through x is $(K_X + B)$ -negative. By the previous paragraph there is a rational curve Γ with $0 < -(K_X + B) \cdot \Gamma \leq 6$ and $R = \mathbb{R}^+[\Gamma]$ — but every such ray lies in $\overline{V}$ by definition of V , a contradiction. Hence $\overline{\text{NA}}(X) = \overline{V} = V$.

The finiteness statements (1)–(2) follow from the boundedness $0 < -(K_X + B) \cdot \Gamma_i \leq 6$ together with a modified-Kähler perturbation argument ([?, Rmk. 4.12–4.13]); see also ?? for the nef supporting class in (2). ■

Remark 4.3 (Höring–Petersen versus Das–Hacon). ?? generalizes ?? from the terminal absolute case ($B = 0$) to dlt pairs, and sharpens the conclusion with the finiteness (1) and the nef supporting class (2). Both proofs reduce to the closedness of V (?), but they differ in how they show every generator of $\overline{\text{NA}}(X)$ lies in $\overline{V}$. Höring–Petersen induct on the ω -degree of curves, applying bend-and-break *directly on $\overline{\text{NE}}(X)$* , and only then transport the conclusion to $\overline{\text{NA}}(X)$ through the three Demailly–Păun generator types $[\hat{\omega}]^2, [\hat{\omega}] \cdot [\hat{S}], [\hat{C}]$ (?). Das–Hacon instead argue *on $\overline{\text{NA}}(X)$ directly*: they take a compact slice, single out an exposed point, and produce a nef supporting class $\alpha = K_X + B + \omega$ with ω Kähler. The latter route is better suited to pairs — it never leaves $\overline{\text{NA}}(X)$ — and it is exactly the form that survives into arbitrary dimension in ??.

5 A relative cone theorem for families of Kähler threefolds

For the Kähler MMP with scaling in dimension four one needs a cone theorem in families of Kähler threefolds. We first fix the (routine but notation-heavy) relative apparatus, then state and completely prove the theorem, following [?, §5, §7–8].

5.1 Relative setup

Throughout, $f: X \rightarrow T$ is a proper surjective morphism from a normal Kähler variety X to a smooth curve T , and $W \subset T$ is a fixed compact subset.

Definition 5.1 (Semi-stable klt pair; [?, Def. 8.1]). Let $B \geq 0$ be an effective $\mathbb{Q}$ -divisor on X . We say $(X, B/T; W)$ is a **semi-stable klt pair** if $(X, X_w + B)$ is plt for every $w \in W$. Equivalently:

- (1) every fibre X_w , $w \in W$, is reduced, irreducible and normal;
- (2) $\text{Supp } B$ contains no fibre X_w (so $B_w := B|_{X_w}$ makes sense);
- (3) $K_X + B$ is $\mathbb{Q}$ -Cartier and (X_w, B_w) is klt for every $w \in W$.

Definition 5.2 (Relative N_1 and $\overline{NA}$; [?, Def. 8.2, p. 57]). Assume every fibre of f is reduced, irreducible and normal. For $u \in U$, an open neighbourhood of W in T , and τ a real closed bidimension- $(1, 1)$ current on X_u , and η a real closed $(1, 1)$ -form on $X_U := f^{-1}U$ with local potentials, set $\tau(\eta) := (\iota_{u,*}\tau)(\eta) = \tau(\eta|_{X_u})$, where $\iota_u: X_u \hookrightarrow X$ is the inclusion. Let $N_1(X/T, W)$ be the real vector space generated by such currents τ on fibres X_w , $w \in W$, modulo $\tau_1 \equiv \tau_2$ iff $\tau_1(\alpha) = \tau_2(\alpha)$ for all $\alpha \in H_{BC}^{1,1}(X_U)$ and some neighbourhood $U \supset W$. Dually, $N^1(X/T, W)$ is the space of classes $\alpha \in H_{BC}^{1,1}(X_U)$ modulo $\alpha_1 \equiv \alpha_2$ iff $[\tau](\alpha_1) = [\tau](\alpha_2)$ for every such τ on every X_w , $w \in W$ (a direct limit over shrinking $U \supset W$). We write $\overline{NA}(X/T, W) \subset N_1(X/T, W)$ for the closed cone generated by classes of positive closed currents, and $[C] \in N_1(X/T, W)$ for the class of a curve $C \subset X_w$, $w \in W$.

A Kähler form ω on X restricts to a Kähler form $\omega_t := \omega|_{X_t}$ on every fibre; we call such ω **relatively Kähler**.

Theorem 5.3 (Relative cone theorem; [?, Theorem 8.5]). Let $f: X \rightarrow T$ be a proper surjective morphism from a Kähler 4-fold X to a curve T with $f_*\mathcal{O}_X = \mathcal{O}_T$, let $W \subset T$ be a compact subset, and let $(X, B/T; W)$ be a semi-stable klt pair. Fix a Kähler form ω on X . Then:

- (a) there are finitely many curve classes $\{C_i\}_{i \in J}$ over W with $0 > (K_X + B) \cdot C_i \geq -6$;
- (b) for each $t \in W$,

$$\overline{NA}(X_t) = \overline{NA}(X_t)_{(K_{X_t} + B_t + \omega_t) \geq 0} + \sum_{i \in J} \mathbb{R}^{\geq 0}[C_i].$$

Suppose moreover that $K_{X_t} + B_t + \omega_t$ is nef for all $t \in W$, with $\omega_t := \omega|_{X_t}$ Kähler, and set $\lambda := \inf\{s \geq 0 \mid K_{X_t} + B_t + s\omega_t \text{ is nef for all } t \in W\}$. If $\lambda > 0$, then there are finitely many classes $\{C_i\}_{i \in I}$, $I \subset J$, over W such that:

- (1) $C_i \subset X_t$ for some $t \in W$, and $\mathbb{R}^{\geq 0}[C_i]$ is a $(K_{X_t} + B_t)$ -negative extremal ray of $\overline{NA}(X_t)$ with $(K_{X_t} + B_t + \lambda\omega_t) \cdot C_i = 0$;
- (2) if $C \subset X_t$ satisfies $(K_{X_t} + B_t + \lambda\omega_t) \cdot C = 0$ for some $t \in W$, then $[C] \equiv \sum_{i \in I} c_i[C_i]$ in $N_1(X/T, W)$ with $c_i \in \mathbb{R}^{\geq 0}$;
- (3) if $\omega \in N^1(X/T, W)$ is general, then $|I| = 1$.

💡 Idea of proof. The proof assembles three ingredients: the threefold cone theorem applied fibre by fibre, a boundedness lemma for families of curves of bounded degree, and an elementary genericity lemma in linear algebra.

Corollary 5.4 (Fibrewise finiteness; [?, Cor. 5.3]). *In the setting of ??, for every $t \in W$ the pair (X_t, B_t) is a compact Kähler threefold dlt pair with $K_{X_t} + B_t$ pseudo-effective, and for the Kähler class ω_t there are only finitely many $(K_{X_t} + B_t + \omega_t)$ -negative extremal rays of $\overline{NA}(X_t)$; each is generated by a rational curve C_i with $0 > (K_{X_t} + B_t) \cdot C_i \geq -6$.*

Proof. By ??, (X_t, B_t) is a klt — in particular dlt — compact Kähler threefold pair for every $t \in W$; that $K_{X_t} + B_t$ is pseudo-effective is part of the standing hypotheses of [?, §5] on the family (guaranteed there by the semi-stability and properness of f over W , which make pseudo-effectivity of $K_{X_t} + B_t$ a closed, hence fibrewise constant, condition on the compact set W). Granting this, the claim is exactly ??(1) applied to (X_t, B_t) with the Kähler class ω_t : finiteness of the $(K_{X_t} + B_t + \omega_t)$ -negative extremal rays, together with ??’s main assertion that each $(K_{X_t} + B_t)$ -negative extremal ray of $\overline{NA}(X_t)$ is generated by a rational curve C_i with $0 < -(K_{X_t} + B_t) \cdot C_i \leq 6$. ■

Lemma 5.5 (Boundedness in families; [?, Lemma 8.4]). *Let $f: X \rightarrow T$, ω relatively Kähler, $W \subset T$ compact, and $M > 0$. Let $\{C_i\}_{i \in I}$ be the set of f -vertical curves with $f(C_i) \subset W$ and $\omega \cdot C_i \leq M$. Then the C_i belong to finitely many families of curves; in particular they represent finitely many classes in $N_1(X/T, W)$.*

Proof. Fix a Kähler form η on X . For each $t \in W$ there is $\epsilon_t > 0$ with $(\omega - \epsilon_t f^* \eta)|_{X_t}$ Kähler on X_t , hence (openness of the Kähler condition) $(\omega - \epsilon_t f^* \eta)|_{X_s}$ Kähler for s near t . By compactness of W we may choose a single $\epsilon > 0$ such that $(\omega - \epsilon f^* \eta)|_{X_t}$ is Kähler for every t in a neighbourhood of W . Then for every i ,

$$\eta \cdot C_i < \frac{1}{\epsilon} \omega \cdot C_i \leq \frac{M}{\epsilon}.$$

Since the relative cycle space (Barlet/Douady space) over W has only finitely many irreducible components of η -degree bounded by M/ϵ , the curves C_i belong to finitely many families. ■

Lemma 5.6 (General and very general Kähler class; [?, Lemma 7.1]). *Let V be a (finite-dimensional) real vector space and $C \subset V$ a cone not contained in any hyperplane. Let $\{C_i\}_{i \in I} \subset V^*$ be a finite (resp. countable) collection with $\omega \cdot C_i := C_i(\omega) > 0$ for all $\omega \in C$, and $C_i \neq \lambda C_j$ for $i \neq j$ and $\lambda \in \mathbb{R}$. Fix $D \in V$. Then there is a finite (resp. countable) union of hyperplanes $\mathcal{H} \subset V$ such that for every $\omega \in C \setminus \mathcal{H}$ and every $t \in \mathbb{R}$, $(D + t\omega) \cdot C_i = 0$ for at most one $i \in I$.*

Proof. Since $C_i \neq \lambda C_j$ for $i \neq j$, the annihilator $\langle C_i, C_j \rangle^\perp \subset V$ has codimension 2. For $i \neq j$ set

$$\mathcal{H}(C_i, C_j) := \{\omega \in C \mid D + t\omega \in \langle C_i, C_j \rangle^\perp \text{ for some } t \in \mathbb{R}\}.$$

If $D + t\omega \in \langle C_i, C_j \rangle^\perp$ then $t\omega \in \langle C_i, C_j \rangle^\perp - D$, so ω lies in the linear span of $\langle C_i, C_j \rangle^\perp - D$, a subspace of codimension ≥ 1 (a hyperplane, since D may shift it off the origin but the span is still proper as $\langle C_i, C_j \rangle^\perp \neq V$); hence $\mathcal{H}(C_i, C_j)$ is contained in a hyperplane. Set $\mathcal{H} := \bigcup_{i \neq j} \mathcal{H}(C_i, C_j)$, a finite (resp. countable) union of hyperplanes. For $\omega \in C \setminus \mathcal{H}$ and any $t \in \mathbb{R}$, $D + t\omega \notin \langle C_i, C_j \rangle^\perp$ for all $i \neq j$; in particular $(D + t\omega) \cdot C_i \neq 0$ for at most one $i \in I$. ■

Proof of ??. *Part (a)–(b).* By ??, for each $t \in T$ there are finitely many $(K_{X_t} + B_t + \omega_t)$ -negative extremal rays, generated by curves C_i , $i \in J_t$, with $0 > (K_{X_t} + B_t) \cdot C_i = (K_X + B) \cdot C_i \geq -6$ (note $(K_X + B) \cdot C_i = (K_{X_t} + B_t) \cdot C_i$ since $C_i \subset X_t$ and $K_X + B$ restricts to $K_{X_t} + B_t$ by the plt/adjunction hypothesis of ??). Let $J := \bigcup_{t \in T} J_t$. Since $\omega \cdot C_i = \omega_t \cdot C_i < -(K_{X_t} + B_t) \cdot C_i \leq 6$ for $i \in J_t$, ?? (with $M = 6$) shows J is finite. This proves (a); statement (b) is ?? restated fibre by fibre, indexed over the finite set J in place of J_t (enlarging the generating set only adds classes already in $\overline{\text{NA}}(X_t)_{(K_{X_t} + B_t + \omega_t) \geq 0}$ for $t' \neq t$, which are non-negative on $K_{X_t} + B_t + \omega_t$ and hence absorbed into the first summand on X_t).

Part (1). Suppose $K_{X_t} + B_t + \omega_t$ is nef for all $t \in W$ and $\lambda > 0$. Set

$$\Lambda := \{t \in W \mid K_{X_t} + B_t + \lambda\omega_t \text{ is nef but not Kähler}\} \subset T.$$

If $\Lambda = \emptyset$, then (arguing as in the proof of ??, using compactness of W to pass from fibrewise Kählerness to a uniform statement over a neighbourhood of W) $K_X + B + \lambda\omega$ would be relatively Kähler over a neighbourhood of W , contradicting the definition of λ as an infimum. So $\Lambda \neq \emptyset$. For $t \in \Lambda$, the class $K_{X_t} + B_t + \lambda\omega_t$ is nef but not Kähler, so by the Kählerness criterion (??) the face

$$F_t := (K_{X_t} + B_t + \lambda\omega_t)^\perp \cap \overline{\text{NA}}(X_t) \neq \{0\}.$$

By part (b) (equivalently ??), F_t is generated by finitely many of the classes $[C_i]$, $i \in J$, each spanning a $(K_{X_t} + B_t)$ -negative extremal ray. Let

$$\Gamma := \{C \subset X_t \mid t \in \Lambda, C \text{ generates a } (K_{X_t} + B_t)\text{-negative extremal ray with } (K_{X_t} + B_t + \lambda\omega_t) \cdot C = 0\}.$$

For $C \in \Gamma$, since $(K_{X_t} + B_t + \lambda\omega_t) \cdot C = 0$,

$$\omega \cdot C = \omega_t \cdot C = -\frac{1}{\lambda}(K_{X_t} + B_t) \cdot C \leq \frac{6}{\lambda}.$$

By ?? (with $M = 6/\lambda$), the curves in Γ belong to finitely many families, hence finitely many numerical classes $\{C_i\}_{i \in I}$, $I \subset J$. This proves (1).

Part (2). Let $C \subset X_t$ satisfy $(K_{X_t} + B_t + \lambda\omega_t) \cdot C = 0$ for some $t \in W$; then $[C] \in F_t$. By part (b) and part (1), F_t is generated by the classes $\{[C_j]\}_{j \in J'}$ for some finite $J' \subset I$ (the ones lying in F_t), so $[C] = \sum_{j \in J'} c_j [C_j]$ with $c_j \in \mathbb{R}_{\geq 0}$ in $H_{\text{BC}}^{1,1}(X_t)$, and hence also, by definition of the relative N_1 -space (??), in $N_1(X/T, W)$.

Part (3). Apply ?? with $V = N^1(X/T, W)$, C the cone of relatively Kähler classes (which is not contained in a hyperplane, since it is open and non-empty), $D = K_X + B$, and $\{C_i\}_{i \in I}$ the finitely many curve classes of part (1) (pairwise non-proportional after discarding repetitions). The lemma produces a finite union of hyperplanes $\mathcal{H} \subset N^1(X/T, W)$ such that for $\omega \in C \setminus \mathcal{H}$ — i.e. for ω general — and $t = \lambda$, $(K_X + B + \lambda\omega) \cdot C_i = 0$ for at most one $i \in I$. Since $\lambda > 0$ was defined so that this vanishing holds for at least one i (part (1)), it holds for exactly one, i.e. $|I| = 1$. ■

Remark 5.7 (Comparison with the threefold and arbitrary-dimensional theorems). ?? is not a new proof of a cone theorem in dimension four: it is the threefold theorem (??), applied fibre by fibre to the Kähler threefolds X_t , $t \in W$, packaged with a uniformity statement across the family. The genuinely new content is ??, which replaces the individual boundedness arguments of ???? (bounded ω -degree on a single X) by a compactness argument over $W \subset T$: it is what lets the finitely-many-rays statement on each fibre assemble into a single finite set J working uniformly over W . ?? is unrelated to positivity; it is used only to force genericity of ω in part (3), and has no analogue in ???? because those sections work with a single fixed variety rather than a family.

6 Hacon–Păun’s approach in arbitrary dimension

Assuming BDPP — which now holds by [?, Theorem 1] — Hacon and Păun prove the cone theorem in all dimensions, ??. The engine is the production of an α -trivial rational curve; the following is the local statement.

Proposition 6.1 ([?, Theorem 0.5, simpler version]). *Assume BDPP. Let (X, ω) be Kähler, $(X, B + \beta)$ generalized klt, and suppose $\alpha = K_X + B + \beta + \omega$ is nef. Then there is an α -trivial rational curve C with*

$$0 > (K_X + B + \beta) \cdot C = \omega \cdot C \geq -2 \dim X.$$

We use the following inputs, all in the generalized-Kähler category: Nakamaye’s theorem, Kawamata’s subadjunction and the canonical bundle formula for generalized pairs ([?]), and Cao–Păun’s positivity of relative adjoint classes.

💡 **Idea of proof.** Locate a maximal-dimensional component Z of the null locus of α , subadjunct to it to see that K_Z is not pseudo-effective, pass to its MRC fibration, and show the generic fibre F — now projective — carries a short α -trivial rational curve via the ordinary cone theorem.

Proof of ??. If α is Kähler there is nothing to prove, so assume α is nef but not Kähler; then $\text{Null}(\alpha)$ has a positive-dimensional component.

Step 1 (null locus as a log canonical center). By Nakamaye's theorem, the restricted non-Kähler locus $E_{nK}^{as}(\alpha)$ coincides with $\text{Null}(\alpha)$, and there is a Kähler current η with weak analytic singularities, $\{\eta\} \in \alpha$, whose Lelong upper-level set equals $E_{nK}^{as}(\alpha)$. Let Z be a maximal-dimensional irreducible component of $\text{Null}(\alpha)$, and let c be the log canonical threshold of $(X, B + \beta_X)$ with respect to η near a general point of Z . Choose a common log resolution $\nu: X' \rightarrow X$ with $K_{X'} + B_{X'} + \beta_{X'} = \nu^*(K_X + B + \beta_X)$, $\beta_{X'}$ nef, and $\nu^*\eta = \eta' + F$ with $F \geq 0$ effective, $\eta' \geq 0$ a positive closed current, and $F + B_{X'}$ SNC. Then Z is a component of $\nu((B_{X'} + cF)^{-1})$ and not contained in $\nu((B_{X'} + cF)^{>1})$. Setting $\beta + c\eta$ with $\eta = \overline{\eta'}$, the analytic subset Z is a generalized log canonical center of the generalized pair $(X, B + c\nu_*F + \beta + c\eta)$.

Step 2 (subadjunction). By Kawamata subadjunction for generalized Kähler pairs, on the normalization Z^ν of Z ,

$$(K_X + B + c\nu_*F + \beta + c\eta)|_{Z^\nu} = K_{Z^\nu} + B_{Z^\nu} + \gamma_{Z^\nu},$$

where $(Z^\nu, B_{Z^\nu} + \gamma_{Z^\nu})$ is a generalized pair, so $B_{Z^\nu} \geq 0$ and γ_{Z^ν} is pseudo-effective. Since $K_X + B + c\nu_*F + \beta + c\eta \equiv (c+1)\alpha - \omega$, restricting gives $K_{Z^\nu} + B_{Z^\nu} + \gamma_{Z^\nu} \equiv (c+1)\alpha_{Z^\nu} - \omega_{Z^\nu}$, where $\alpha_{Z^\nu} = \alpha|_{Z^\nu}$, $\omega_{Z^\nu} = \omega|_{Z^\nu}$. With $k := \nu_{\text{num}}(\alpha_{Z^\nu}) < \dim Z$ (so $\alpha_{Z^\nu}^k \neq 0$, $\alpha_{Z^\nu}^{k+1} \equiv 0$),

$$(K_{Z^\nu} + B_{Z^\nu} + \gamma_{Z^\nu}) \cdot \alpha_{Z^\nu}^k \cdot \omega_{Z^\nu}^{\dim Z - k - 1} = -\alpha_{Z^\nu}^k \cdot \omega_{Z^\nu}^{\dim Z - k} < 0,$$

the strict inequality because α_{Z^ν} is nef and $\alpha_{Z^\nu}^k \neq 0$.

Step 3 (MRC fibration). Since $B_{Z^\nu} \geq 0$ and γ_{Z^ν} is pseudo-effective, the displayed negativity forces K_{Z^ν} (hence $K_{Z'}$ for any resolution $Z' \rightarrow Z^\nu$) to be non-pseudo-effective. Consider the MRC fibration $Z' \rightarrow Y$, non-trivial as $K_{Z'}$ is not pseudo-effective; after passing to a higher model it is a morphism with general fibre F . As F is rationally connected, $h^2(\mathcal{O}_F) = 0$, so F is projective, and F is rationally connected.

Step 4 (α_F is not big). For any $\epsilon > 0$ and $t > 0$,

$$(K_{Z'} + B_{Z'} + \gamma_{Z'} + (1 - \epsilon)\omega_{Z'} + t\alpha_{Z'}) \cdot \alpha_{Z'}^k \cdot \omega_{Z'}^{\dim Z - k - 1} = -\epsilon \alpha_{Z'}^k \cdot \omega_{Z'}^{\dim Z - k} < 0,$$

and since $\gamma_{Z'}$ is pseudo-effective and $B_{Z'}^{<0}$ is $(Z' \rightarrow Z^\nu)$ -exceptional, $K_{Z'} + (1 - \epsilon)\omega_{Z'} + t\alpha_{Z'}$ is not pseudo-effective for any $t > 0$. As Y is not uniruled, K_Y is pseudo-effective; applying Cao–Păun positivity of the relative adjoint class to $Z' \rightarrow Y$ shows $K_F + (1 - \epsilon)\omega_F + t\alpha_F$ is not pseudo-effective for any $t > 0$. In particular α_F is not big (else it would be pseudo-effective for $t \gg 0$).

Step 5 (fibre-type contraction). By the projective cone theorem applied to $(F, (1 - \epsilon)\omega_F)$, there are finitely many $(K_F + (1 - \epsilon)\omega_F)$ -negative extremal rays, and (comparing with α_F , which is nef) a non-empty finite subcollection of them is α_F -trivial. Let $\eta: F \rightarrow \bar{F}$ contract such a face; then $\alpha_F = \eta^*\alpha_{\bar{F}}$ with $\alpha_{\bar{F}}$ ample. If η were birational, α_F would be big — contradiction. Hence η is of fibre type, and F is covered by α_F -trivial rational curves C .

Step 6 (length). By bend-and-break we may take $0 < -K_F \cdot C \leq 2 \dim F$, hence $(1 - \epsilon)\omega_F \cdot C < 2 \dim F \leq 2 \dim X$. Thus Z' (and so Z) is covered by α -trivial rational curves C with

$0 < \omega \cdot C < \frac{2}{1-\epsilon} \dim X$. These lie in finitely many numerical classes, so letting $\epsilon \rightarrow 0$ we may assume $0 < \omega \cdot C \leq 2 \dim X$. Finally, by α -triviality,

$$0 < -(K_X + B + \beta_X) \cdot C = \omega \cdot C \leq 2 \dim X. \quad \blacksquare$$

💡 Idea of proof. By ??, every nef adjoint class $\alpha = \Theta + \omega$ that is not Kähler already carries a short α -trivial rational curve; feeding this into the exposed-point argument of ?? verbatim gives the cone theorem in dimension n .

Proof of ??. Write $\Theta := K_X + B + \beta_X$. By ??, every nef adjoint class $\alpha = \Theta + \omega$ (ω Kähler) that is not Kähler carries an α -trivial rational curve C with $0 < -\Theta \cdot C = \omega \cdot C \leq 2n$. This is the Mori bend-and-break input; the rest of the argument is the exposed-point argument of ??.

Extremal rays are short rational curves. Let K be a compact convex slice of $\overline{\text{NA}}(X)$ of Euclidean dimension $\dim_{\mathbb{R}} H_{\text{BC}}^{1,1}(X) - 1$, and R the ray of an exposed point $\{x\} = K \cap \alpha^\perp$, where α is nef (??) with $K \subset \alpha_{\geq 0}$. If $\Theta \cdot R < 0$, then for $0 < a \ll 1$ the class $\beta := \alpha - a\Theta$ is strictly positive on $\overline{\text{NA}}(X) \setminus \{0\}$, hence Kähler; rescaling, $\alpha \equiv \Theta + \omega$ with ω Kähler, so the previous paragraph applies and $R = \mathbb{R}^+[\Gamma]$ for a rational curve Γ with $0 < -\Theta \cdot \Gamma \leq 2n$. A Douady space argument (as X is Kähler) shows there are at most countably many such rays $\{\mathbb{R}^+[\Gamma_i]\}_{i \in I}$.

Closedness and conclusion. Put $V := \overline{\text{NA}}(X)_{\Theta \geq 0} + \sum_{i \in I} \mathbb{R}^+[\Gamma_i]$. By ?? (generalized-lc form, with $0 < -\Theta \cdot \Gamma_i \leq 2n$), V is closed, so it suffices to prove $\overline{\text{NA}}(X) = \overline{V}$. If not, since $\overline{V}$ is convex and $\overline{\text{NA}}(X)$ is generated by K , there is an exposed point $x \in K \setminus \overline{V}$; as $V \supset \overline{\text{NA}}(X)_{\Theta \geq 0}$, its ray R is Θ -negative, hence $R = \mathbb{R}^+[\Gamma]$ for a rational curve with $0 < -\Theta \cdot \Gamma \leq 2n$ by the previous step. But every such ray lies in $\overline{V}$ by definition of V — a contradiction. Thus

$$\overline{\text{NA}}(X) = \overline{\text{NA}}(X)_{\Theta \geq 0} + \sum_{i \in I} \mathbb{R}^+[\Gamma_i], \quad 0 < -\Theta \cdot \Gamma_i \leq 2n.$$

Finally, if $B + \beta_X$ (or Θ) is big, then after a modified-Kähler perturbation ω' every Θ -negative Γ_i has bounded ω' -degree while ω' is strictly positive on $\overline{\text{NA}}(X) \setminus \{0\}$; hence the classes $[\Gamma_i]$ cannot accumulate, and I is finite. $\blacksquare$

Remark 6.2 (What changes in higher dimension). The arbitrary-dimensional argument (??) replaces the surface-specific input of ?? — the Zariski decomposition of a pseudo-effective K_X on a threefold, and Mori bend-and-break on the resulting uniruled surfaces — by two dimension-free tools. Nakamaye’s theorem describes the null locus of the nef-but-not-Kähler class α as the Lelong set of a Kähler current, and BDPP (now Ou’s theorem [?]) applied to the MRC fibration of a maximal null-locus component produces the covering rational curves. The subadjunction of Step 2 plays the role that the surface adjunction $K_{S_j} \cdot C < K_X \cdot C$ did in dimension three. This is precisely why

the theorem is conditional *only* on BDPP: once BDPP holds in the relevant dimension, the bend-and-break input — a short α -trivial rational curve — is available in every dimension, and the exposed-point argument of ?? then finishes verbatim.

7 Applications of the cone theorem

The cone theorem is the first pillar of the (Kähler) minimal model program: it locates the $(K_X + B)$ -negative extremal rays, which are exactly the rays one contracts. We record the principal consequences.

7.1 The contraction theorem and the elementary contractions

Each negative extremal ray can be contracted in the Kähler category, and the contraction falls into one of three types.

Theorem 7.1 (Contraction theorem; [?, Thm. 1.3, Cor. 7.7–7.9, Thm. 7.12], [?]). *Let X be a normal $\mathbb{Q}$ -factorial compact Kähler threefold with terminal singularities and K_X pseudo-effective (or, more generally, (X, B) a $\mathbb{Q}$ -factorial compact Kähler 3-fold dlt pair with $K_X + B$ pseudo-effective). Let $R = \mathbb{R}^+[\Gamma]$ be a $(K_X + B)$ -negative extremal ray of $\overline{\text{NA}}(X)$, with supporting nef class $\alpha = K_X + B + \eta$ as in ??(2). Then the contraction $\text{contr}_R: X \rightarrow Y$ exists in the Kähler category, and exactly one of the following holds:*

- (i) (divisorial) contr_R is bimeromorphic and contracts a prime divisor S — to a point if the nef dimension $n(\alpha) = 0$, to a curve if $n(\alpha) = 1$;
- (ii) (flipping / small) contr_R is bimeromorphic and small, contracting only finitely many curves; the R -flip $X \dashrightarrow X^+$ then exists;
- (iii) (fibre type / Mori fibre space) $\dim Y < \dim X$ and $-(K_X + B)$ is contr_R -ample, exhibiting $X \rightarrow Y$ as a Mori fibre space.

The nef supporting class α of ??(2) is precisely the input for building contr_R : one shows α is semi-ample (a Kähler analogue of the base-point-free theorem, through a nef-and-big Kählerness criterion), and contr_R is the associated fibration. The fibre-type case realises X as uniruled; for Kähler 3-folds whose MRC fibration has surface base this is [?, Thm. 3.15].

7.2 One step of the Kähler MMP

Let (X, B) be a $\mathbb{Q}$ -factorial compact Kähler 3-fold dlt pair. If $K_X + B$ is nef, X is already a minimal model. Otherwise ?? produces a $(K_X + B)$ -negative extremal ray R , and ?? contracts it: a divisorial contraction or a flip replaces (X, B) by a new $\mathbb{Q}$ -factorial dlt pair with smaller Picard number or milder discrepancies, while a fibre-type contraction ends the program with a Mori fibre space. Iterating and invoking termination runs the Kähler

MMP in dimension three [?, ?]. The finiteness in ??(1) is exactly what makes MMP *with scaling* well defined: for a fixed Kähler ω only finitely many rays are $(K_X + B + \omega)$ -negative, so the nef threshold is attained and the scaling proceeds.

7.3 Families and higher dimension

The relative cone theorem (??) plays the same role for the Kähler MMP *with scaling in families* and is a key ingredient in the 4-dimensional Kähler MMP [?]. In arbitrary dimension, ?? supplies the extremal rays for generalized klt pairs, so that — granting the existence of the corresponding contractions and flips — the same one-step picture is expected to run the generalized Kähler MMP.

Key takeaways

- The cone theorem always has the same shape: countably many bounded-length rational curves generate the negative part of the relevant cone, with the nef/pseudo-effective hypothesis on $K_X + B$ supplying either a Zariski decomposition or a well-behaved null locus to run bend-and-break.
- The generalized Mori cone $\overline{NA}(X)$, not the classical $\overline{NE}(X)$, is the right object in the Kähler category (??).
- Each negative extremal ray contracts to a divisorial, flipping, or fibre-type morphism (??); iterating this is one step of the Kähler MMP, and the finiteness clause of the cone theorem is exactly what makes MMP with scaling well posed.
- Passing to families (??) or arbitrary dimension (??) reuses the threefold argument verbatim, once a boundedness lemma over the family, resp. BDPP, is granted.

8 Further discussions

- (1) The optimal bound for Kawamata's length of extremal rays has recently been established; the sharp constant improves the $2n$ (resp. 6 in dimension three) above.
- (2) Fujino generalised the cone theorem to the log canonical and semi-log canonical cases.
- (3) Corti–Lazić gave an alternative approach to the cone theorem via finite generation.